

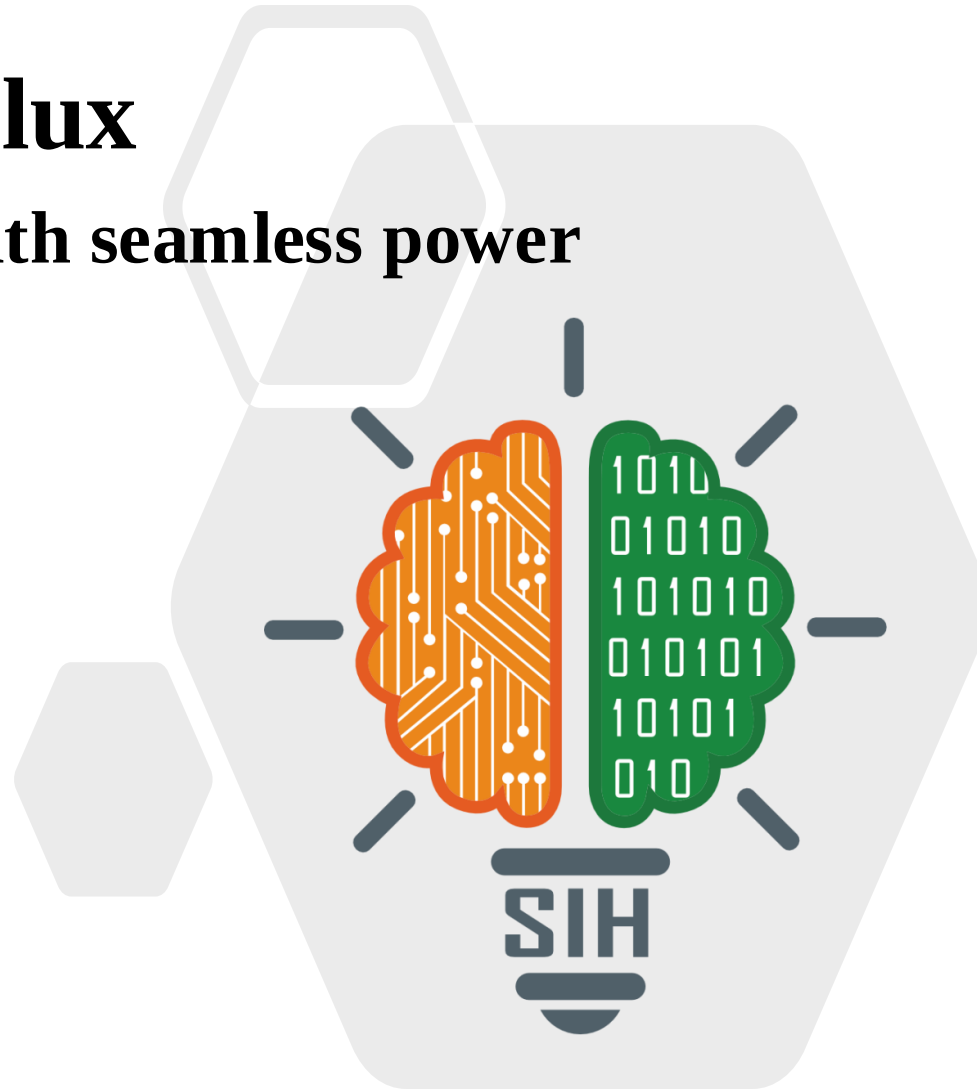
SMART INDIA HACKATHON 2024



InFlux

Charging ahead with seamless power

- **Problem Statement ID- SIH1595**
- **Problem Statement Title- InFlux**
- **Theme- Smart Vehicle**
- **PS Category- Software**
- **Team ID- MITADTSW337**
- **Team Name- C2 Innovatives**



InFlux

Charging ahead with seamless power

❖ Proposed Solution

- Provide a centralized website for locating EV charging stations.
- Collaborate with all major EV charger brands.
- Offer a unified wallet for easy payments across different chargers.



C2 Innovatives

SOFTWARES

FRONTEND

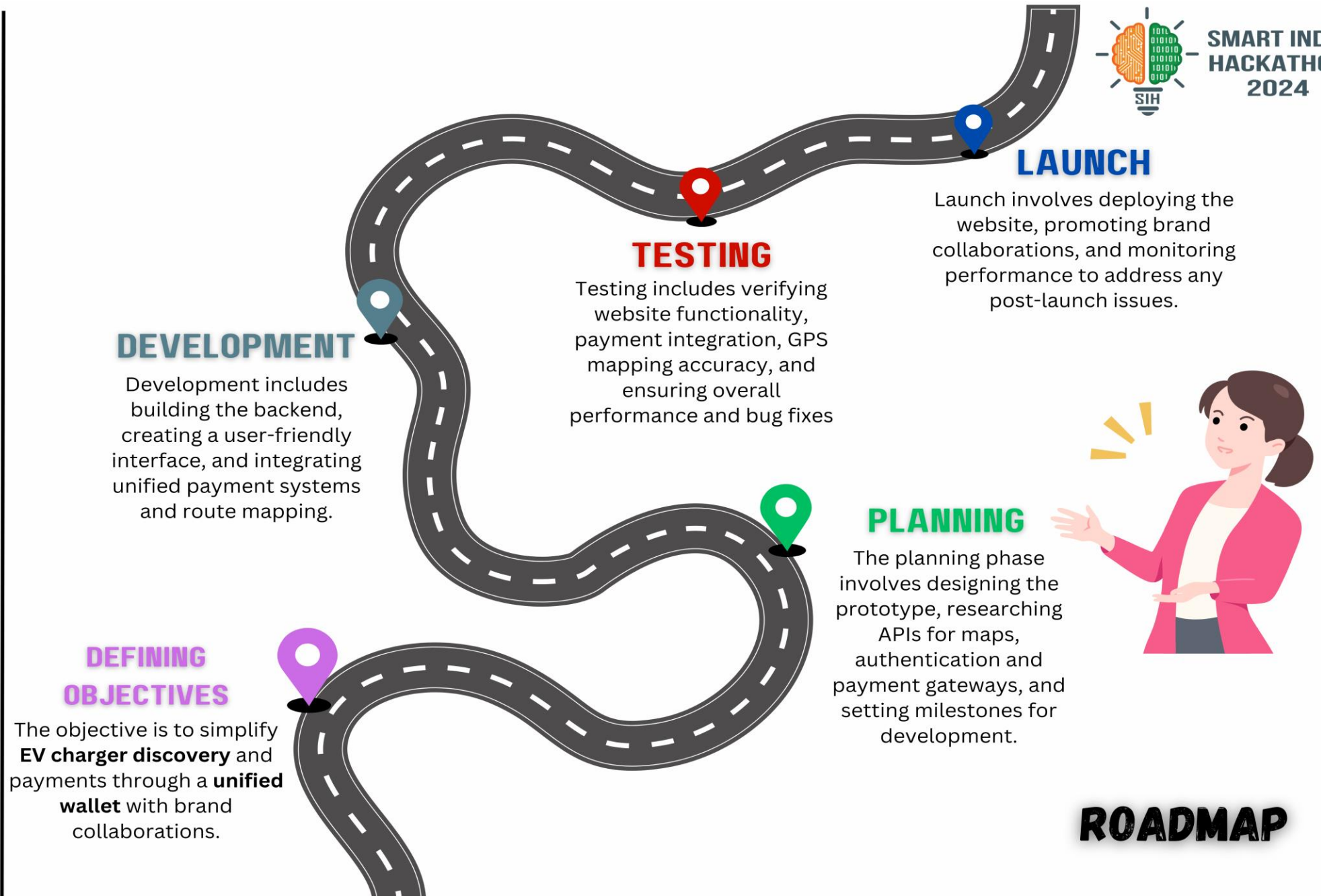
- React.js
- HTML / CSS
- BootStrap

BACKEND

- Node.js
- Express.js
- API Calls (Axios, Fetch)
- Google Maps API
- Payment Gateway API

DATABASES

- MONGODB
- MY SQL



Challenge 1
Demand vs Infrastructure

Rapid EV adoption may outpace charging infrastructure development.



Solution

- Growing demand for **web-based EV charging solutions** with market growth.

Challenge 2
Brand Integration:

Managing multiple charger APIs can lead to compatibility and data consistency issues.



Solution

- Collaborating with different brands to reduces infrastructure challenges.

Challenge 3
Payment Security:

Users may hesitate due to concerns about online payment safety.



Solution

- Secure payment systems ensures user trust and adoptions.

Challenge 4
Scalability:

Ensuring smooth platform growth and future mobile compatibility can be challenging.



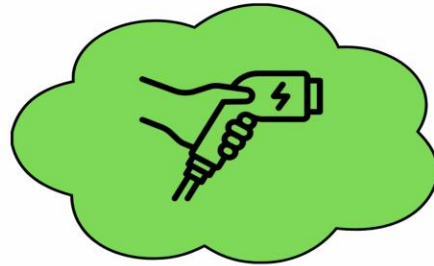
Solution

- Scalable for future additions like **mobile app compatibility**.

IMPACT AND BENEFITS

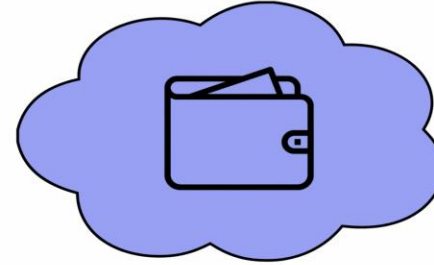
01 | EFFICIENT CHARGER DISCOVERY

The website simplifies locating EV charging stations, providing users with real-time station data.



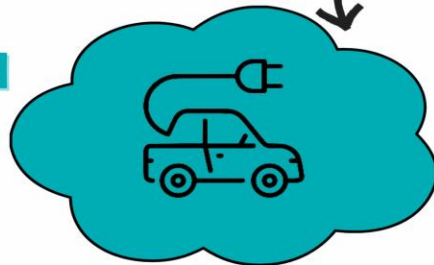
02 | ENHANCED PAYMENT CONVENIENCE

The unified wallet streamlines payments, boosting user convenience and accelerating transaction speed.



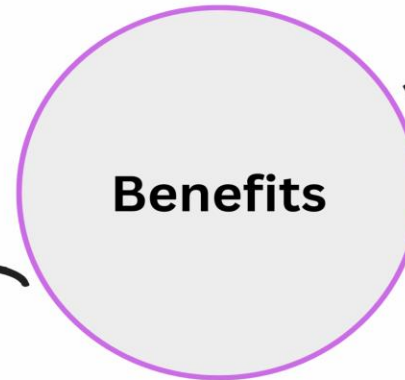
03 | PROMOTING SUSTAINABLE TRANSPORTATION

By supporting EV adoption, the platform contributes to the growth of green transportation and environmental sustainability.



04 | REVENUE GROWTH FOR CHARGER PROVIDERS

Collaboration with EV charger brands creates new revenue streams and business opportunities for providers.



RESEARCH AND REFERENCES

- ❑ Acharige, S. S., Haque, M. E., Arif, M. T., Hosseinzadeh, N., Hasan, K. N., & Oo, A. M. T. (2023). Review of electric vehicle charging technologies, standards, architectures, and converter configurations. *IEEE Access*, 11, 41218-41255.
- ❑ Flocea, R., Hîncu, A., Robu, A., Senocico, S., Traciu, A., Remus, B. M., ... & Filote, C. (2022). Electric vehicle smart charging reservation algorithm. *Sensors*, 22(8), 2834.
- ❑ Singh, V., Singh, V., & Vaibhav, S. (2021). Analysis of electric vehicle trends, development and policies in India. *Case Studies on Transport Policy*, 9(3), 1180-1197.